

$$6.1) 12)(b) \exists k \in \mathbb{Z} (9k = 4^n + 6n - 1) \rightarrow \exists k \in \mathbb{Z} (9k = 4^{n+1} + 6(n+1) - 1)$$

$$4^{n+1} + 6(n+1) - 1 = 4^n \cdot 4 + 6n + 6 - 1$$

$$= 4^n \cdot 4 + 6n + 5 + \underline{18n - 18n + 4 - 4}$$

$$= 4^n \cdot 4 + 24n - 4 - 18n + 9$$

$$= 4(4^n + 6n - 1) - 18n + 9$$

$$= 4(9k) - 9(2n - 1)$$

$$= 9(4k - 2n + 1)$$

so $l = 4k - 2n + 1$. Hence Proved!

$$6.1) 13) \text{ Note that } a^{n+1} - b^{n+1} = a(a^n - b^n) + ?$$

$$= a^{n+1} - ab^n$$

so we need to ^{add} ~~(subtract)~~ ab^n & ^(add) ~~(subtract)~~ b^{n+1} but $(b^{n+1} - ab^n) = b^n(a - b)$

$$= b^n(a - b)$$

$$\text{so } a^{n+1} - b^{n+1} = a(a^n - b^n) + b^n(a - b)$$

Now to use induction on n . Base case is $n=0$, $a-b \mid a^0 - b^0 = 1 - 1 = 0$

Induction step, supp $(a-b) \mid (a^n - b^n)$ so $\exists k \in \mathbb{Z} [(a-b)k = a^n - b^n] - \textcircled{1}$

$$\text{Then } a^{n+1} - b^{n+1} = a(a^n - b^n) + b^n(a - b)$$

$$= a(a-b)k + b^n(a-b)$$

$$= (a-b)(ak + b^n)$$

$$\text{so } (a-b) \mid a^{n+1} - b^{n+1}$$

6.1) 14) let $a, b \in \mathbb{Z}$ be arbitrary. We will use Induction on n .

Base case: $n=0$ then $(a+b) | (a+b)$ clearly.

Induction step: Supp $(a+b) | (a^{2^{n+1}}, b^{2^{n+1}}) \rightarrow \exists k \in \mathbb{Z} [(a+b)k = a^{2^{n+1}} + b^{2^{n+1}}]$

$$\begin{aligned} \text{Then } a^{2^{(n+1)+1}} + b^{2^{(n+1)+1}} &= a^{2^{n+2}} + b^{2^{n+2}} = a^2 a^{2^{n+1}} + b^2 b^{2^{n+1}} \\ &= a^2 (a^{2^{n+1}} + b^{2^{n+1}}) + b^{2^{n+1}} (a^2 - b^2) \\ &= a^2 (a+b)k + b^{2^{n+1}} (a+b)(a-b) \\ &= (a+b) [a^2 k + b^{2^{n+1}} (a-b)] \end{aligned}$$

so $(a+b) | a^{2^{(n+1)+1}} + b^{2^{(n+1)+1}}$.

6.1) 17) Induction step:

$$2 \cdot 2^1 + 3 \cdot 2^2 + 4 \cdot 2^3 + \dots + (n+1)2^n + (n+2)2^{n+1}$$

$$= n2^{n+1} + (n+2)2^{n+1} = n2^{n+1} + n2^{n+1} + 2^{n+1} \cdot 2$$

$$= 2n2^{n+1} + 2 \cdot 2^{n+1} = n(2^{n+2}) + 2^{n+2} = (n+1)2^{n+2}.$$

6.1) 20) $a \in \mathbb{R}, b \in \mathbb{R}, 0 < a < b \rightarrow$ ①

skipping Base cases.

(a) $\forall n \geq 1, 0 < a^n < b^n$ ② $\longrightarrow 0 < a^{n+1} < b^{n+1}$

As $0 < a$ & $0 < a^n$ so $0 < a \cdot a^n = a^{n+1}$

Also as $a < b$ & $a^n < b^n$ so $a \cdot a^n < b \cdot b^n \rightarrow a^{n+1} < b^{n+1}$

(b) $\forall n \geq 2, 0 < \sqrt[n]{a} < \sqrt[n]{b} \longrightarrow 0 < \sqrt[n]{a} < \sqrt[n]{b}$

To avoid complex fractions, let $x = \sqrt[n]{a}$ & $y = \sqrt[n]{b}$ ③

We can prove $0 < x < y$

By ③, $x^{n+1} = a$ & $y^{n+1} = b$ ④

To see that x & y are +ve,

Case 1) n is odd. If $x < 0$ then as n is odd, $x^n < 0$

by exercise 19 but $x^n = a$ so $a < 0$ is contradiction to ①.

Case 2) n is even, then $x^n = x^{2k}$ for $k \geq 1$, so $(x^k)^2 = a$

Then x can be +ve or -ve but $\sqrt[n]{a}$ is by definition, a

For $n \geq 1$, some reasoning applies if n is odd, $n+1$ is even

positive root. similar for $\sqrt[n]{b}$. so $0 < x$ & $0 < y$.

To see $x < y$. we will use proof by contradiction.

supp $x \geq y$ — (5)

Case 1) $x = y$. Case 2) $x > y$.

We can easily combine both cases. By (1), as $a < b$ so $x^{n+1} < y^{n+1}$. But along with (5), this is contradiction to part (a).

(c) $\forall n \geq 1, ab^n + ba^n < a^{n+1} + b^{n+1} \rightarrow ab^{n+1} + ba^{n+1} < a^{n+2} + b^{n+2}$.

Equivalent to $a^{n+2} + b^{n+2} - ab^{n+1} - ba^{n+1} > 0$

$\Leftrightarrow a(a^{n+1} - b^{n+1}) - b(a^{n+1} - b^{n+1}) > 0$

$\Leftrightarrow (a-b)(a^{n+1} - b^{n+1}) > 0$.

By (1), $0 < a < b$ so $a - b < 0$.

And by part (a), $a^{n+1} < b^{n+1}$, so $a^{n+1} - b^{n+1} < 0$.

so $(a-b)(a^{n+1} - b^{n+1}) > 0$ $\therefore$ Product of 2 -ve numbers = +ve

(d) $\forall n \geq 2, \left(\frac{a+b}{2}\right)^n < \frac{a^n + b^n}{2} \xrightarrow{(6)} \left(\frac{a+b}{2}\right)^{n+1} < \frac{a^{n+1} + b^{n+1}}{2}$

Multiplying (6) by $\frac{a+b}{2}$, we get

$\left(\frac{a+b}{2}\right)^{n+1} < \left(\frac{a^n + b^n}{2}\right)\left(\frac{a+b}{2}\right) = \frac{a^{n+1} + ab^n + ba^n + b^{n+1}}{4}$ — (7)

Note by part (c), $ab^n + ba^n < a^{n+1} + b^{n+1}$

so $ab^n + ba^n + a^{n+1} + b^{n+1} < 2a^{n+1} + 2b^{n+1}$

($\therefore$ (7) Adding $a^{n+1} + b^{n+1}$ & 2 dividing by 2) $\frac{2}{2}$
 $= 2(a^{n+1} + b^{n+1})$
 $= a^{n+1} + b^{n+1}$

so $ab^n + ba^n + a^{n+1} + b^{n+1} < a^{n+1} + b^{n+1}$

dividing by 2, $ab^n + ba^n + a^{n+1} + b^{n+1} < a^{n+1} + b^{n+1}$ — (8)

this, combining (7) & (8), 4 proves the goal.